

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 5

6 Computer Networking (awm22)

Networks(3):
IPv6

(a) For IPv4 ISPs, each domestic installation typically gets a /32 network. You have a complicated configuration requiring NAT and multiple IPv4 subnets.

(i) Why would an IPv6 based provider allocate four /64 networks for your premises when each /64 represents 2^{64} addresses? [2 marks]

Answer: (Officially) IPv6 does not do NAT; so a single /64 (which provides 2^{64} addresses) allocation can not be split into smaller blocks without breaking ZeroConf. Officially a /64 is the address allocation for a single IPv6 network.

Four /64 are given to permit a home owner to separate devices that ought not share the same network: IoT devices, home server, DMZ devices, guest WiFi network, and so forth.

(ii) A colleague has IPv6 with another provider; they only allow one /64 for each domestic installation. In the past your colleague has used a NAT and many IPv4 private address blocks, but keenly adopted IPv6 permitting them to upgrade their home network. They are now using blocks of the allocated /64 and a router in their home to interconnect the subnets.

Not everything is working as they hoped; for example, sometimes IoT devices can't connect to the Internet to update and your colleague can not connect to their front-door camera when at work.

Explain what sort of problems your colleague may face along with methods by which they could verify the root cause. [6 marks]

Answer: No NAT.

The /64 will be routable but this depends greatly on each device being able to configure addresses correctly.

The /64 allocation should not be split into subnets as it can break the ZeroConf mechanic of IPv6; for example this can mean a device has a non routable local address and did not complete acquiring a globally routable address.

This means that depending on which network the device is attached, or worse, which network it completes its address discovery on, the device may or may not have a globally routable address.

Diagnostics will mean local printer might work from local laptop.

Laptop on the globally routable address block will work ok.

The router will inform all the hosts about its acting as a router.

All in all a bit of a mess.

If the colleague uses PLA for subnets everything gets worse. PLA are not routable onto the 'Internet', nor will these systems every be able to be reachable from the public internet.

Diagnostics would indicate addresses taken from the ISP /64 are all routable; while the PLA are not.

If there is a local 6NAT6 this would suffer all the typical problems of a NAT (not routable-to).

The local router will allow routing between the local subnets - which can further hide / complicate the diagnostics.

- (iii) Explain to your colleague why you might not be able to *lend them* one of your /64 allocations, even though the /64 blocks (provided to you by your ISP) are each globally routable addresses. [2 marks]

Answer: Ah, where to start....

critically, the address blocks are for a particular premises/customer.

The address is part of the ISP routing, it will advertise a netblock that includes your four /64 (aka a /62) - along with many other IPv6 addresses. thus the global IPv6 network will keep sending packets bound for your (four /64) addresses irrespective of your colleague using that /64 for their home network.

It is possible the colleague's home network will work a little - the lent /64 will behave a little like a private link address; but behaviour would be indeterminate.

CSMA/CA,
CSMA, physical
layer,
multiplexing

- (b) A local area network may carry several different LANs simultaneously; such a network would be designated for known sets of HomePlug devices.

Describe a physical line coding approach for the HomePlug devices that: allows two or more simultaneous virtual local area networks to fairly share the same physical channel, but does not permit trivial interception of network traffic.

Outline your approach along with its benefits and drawbacks, comparing it with the simplest use of VLAN tags in Ethernet. [10 marks]

Answer: I'm expecting students to describe CDMA although any orthogonal coding scheme will work.

CDMA provides a relatively low effective data-rate over a base channel.

Decent selection of the keys provides a great encryption

Candidates may need to define fairness; thus we might have:

Classical CSMA / CA but with VLAN enciphered within frames - a crypto channel for each VLAN would be a solution, but this is not a change to the physical line coding.

Perhaps a more elegant approach might be CDMA on top of CSMA/CD where each network HomePlug device knows the encryption key common to the VLAN.

Simply using different physical pair keys will provide most of the properties but the fairness is only true "on average" aka probabilistically; there is no specific...

4 marks for proposing a technology that gives a fair approach

2 marks recognising the HomePlug device must encipher

2 marks for a schema that is not trivial to decipher (eg not FDMA or STDMA)

2 marks for recognising the advantages/drawbacks of the scheme proposed.
